23AID111

PROJECT SUBMISSION

# GOODS VEHICLE MANAGEMENT AND BOOKING SYSTEM

GROUP 12

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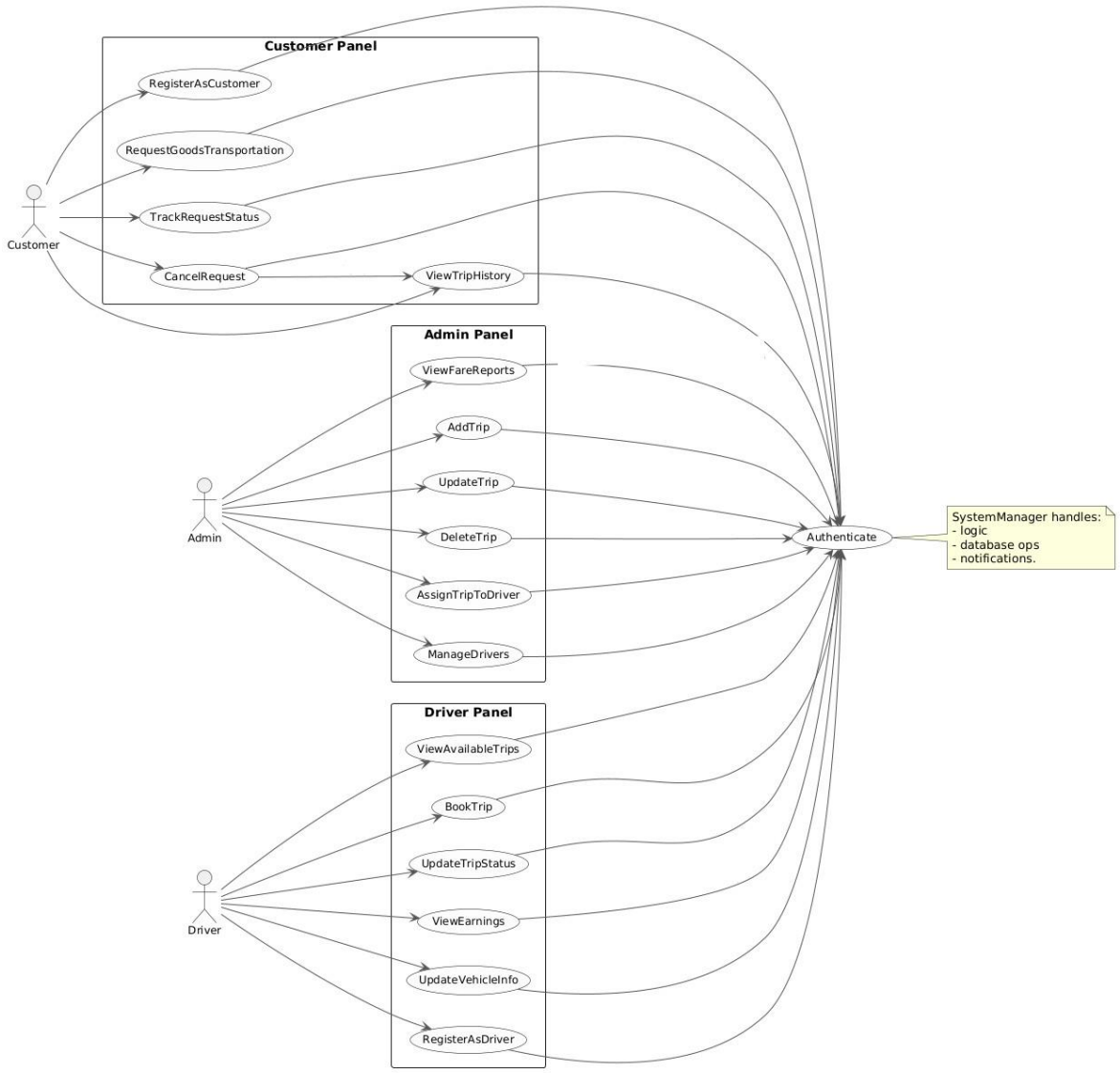
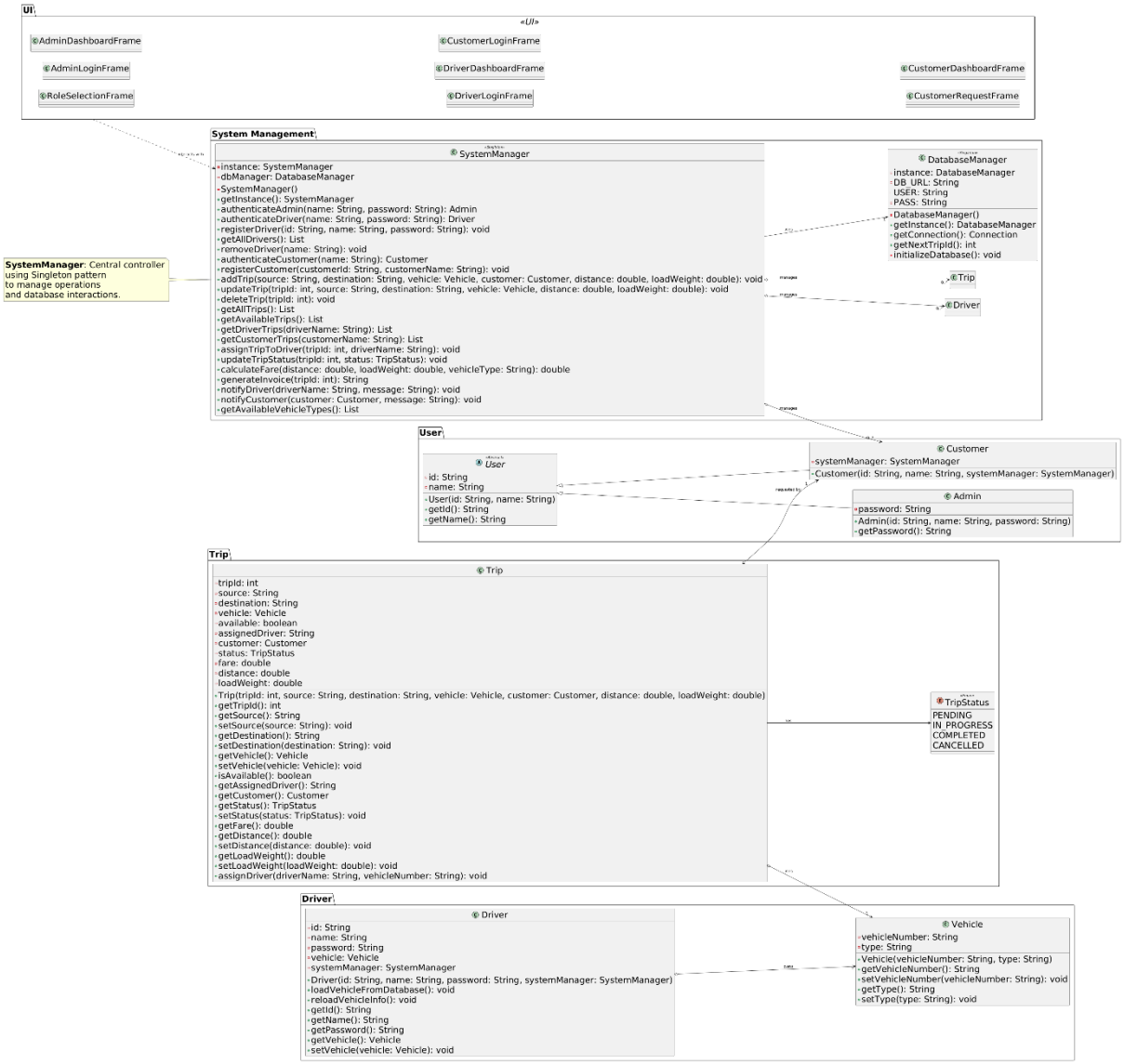
## ABSTRACT

The Goods Vehicle Management and Booking System (GVBMS) is a Java-based application designed to streamline the logistics of goods transportation by facilitating efficient management and booking of vehicles. Built on a MySQL database, the system supports three primary user roles: Admin, Driver, and Customer. Admins oversee operations, including trip management, driver registration, and fare reporting. Drivers can view and book available trips, update trip statuses, and manage vehicle information. Customers can request transportation services, track their requests, and cancel bookings. The system employs a Singleton pattern for centralized management, JDBC for database connectivity, and Swing for a user-friendly graphical interface. Key functionalities include trip scheduling, fare calculation based on distance, load weight, and vehicle type, and real-time notifications via dialog boxes. GVBMS ensures robust error handling and data integrity, making it a reliable solution for small to medium-scale logistics operations.

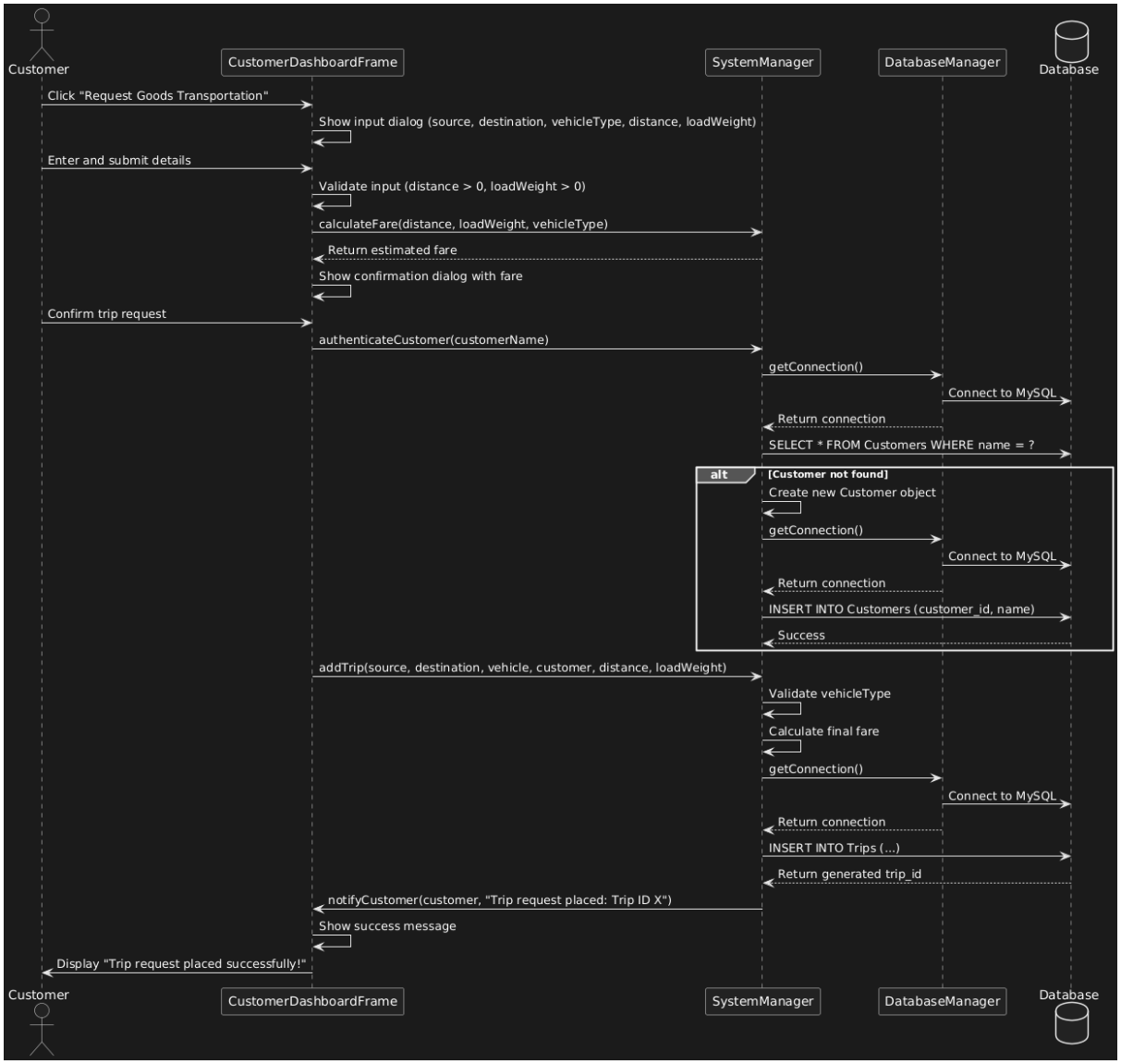
# USE CASE DIAGRAM



# CLASS DIAGRAM

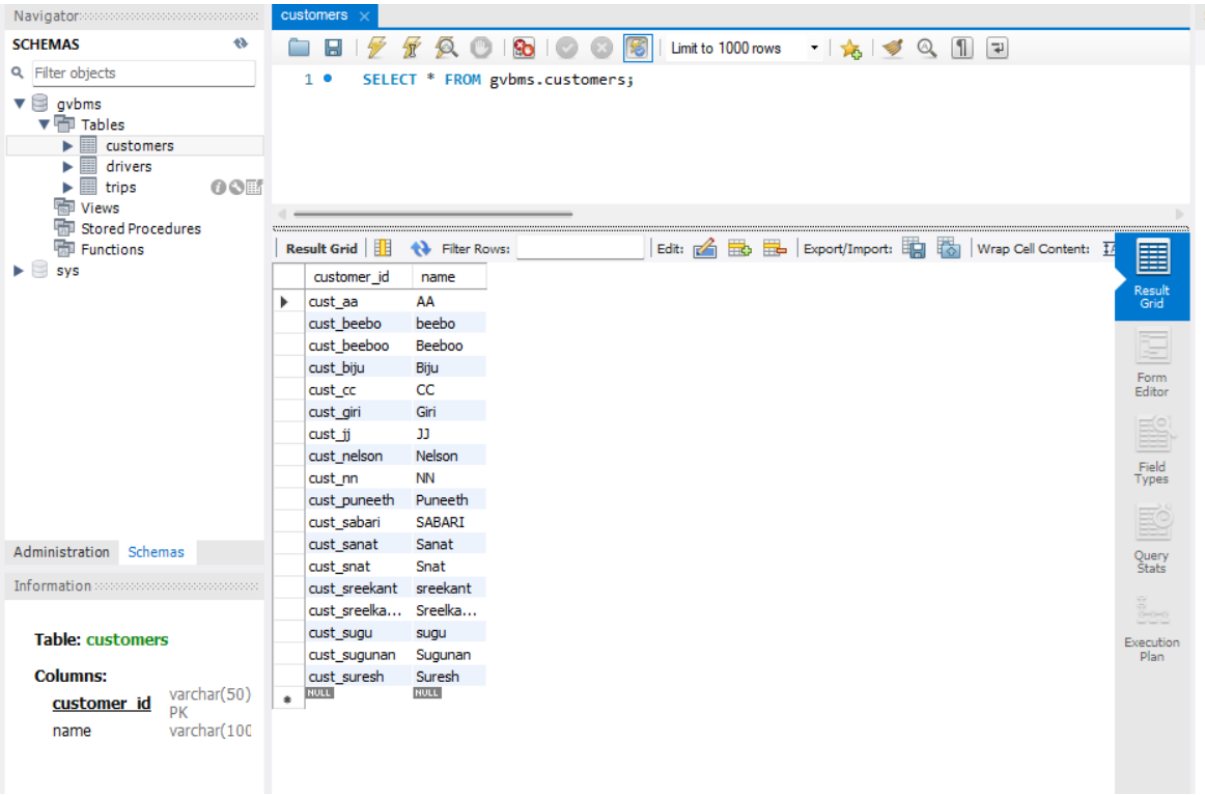


# SEQUENCE DIAGRAM

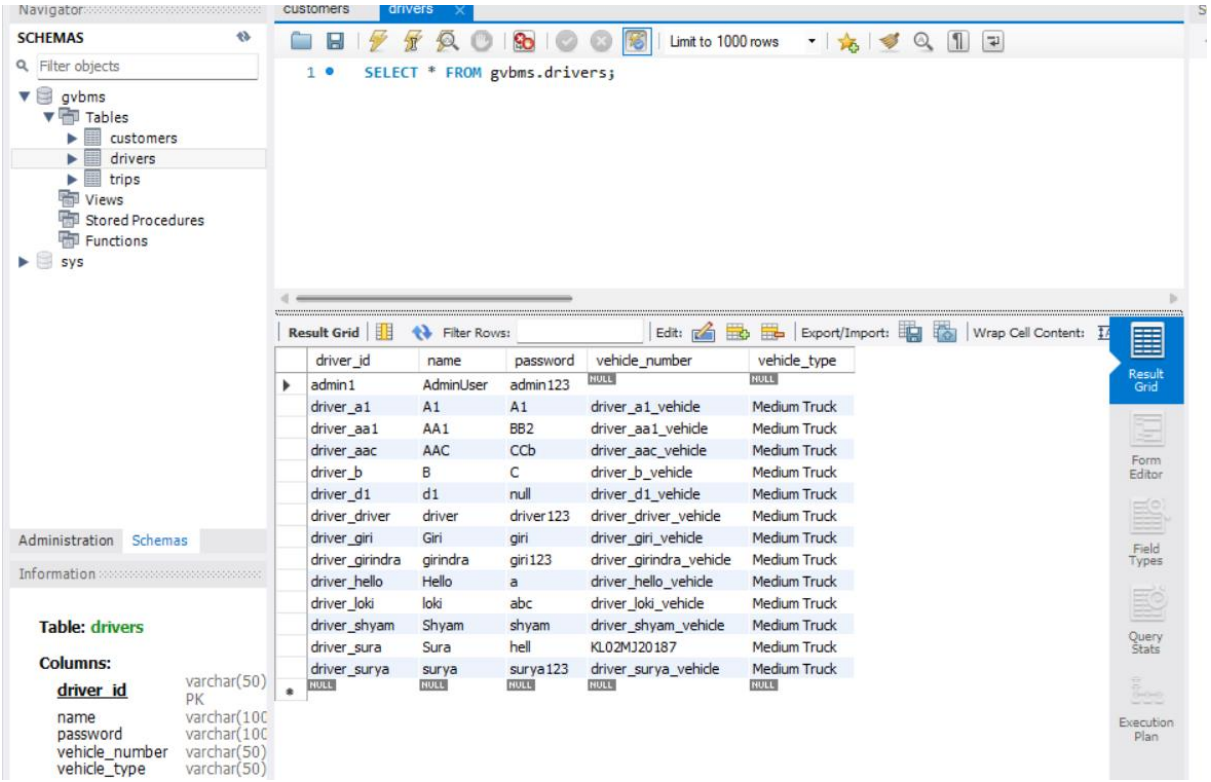


# SQL TABLES

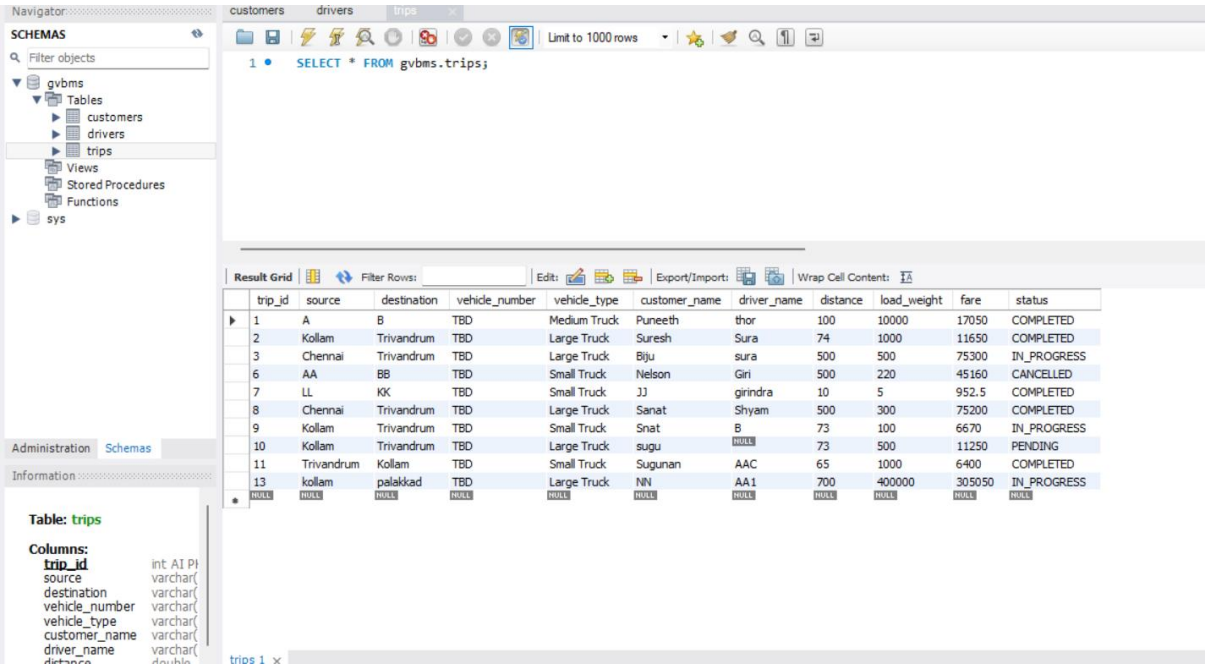
## Customer table



Drivers table



Trips table




| trip_id | source | destination | vehicle_number | vehicle_type | customer_name | driver_name | distance | load_weight | fare | status |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | A | B | TBD | Medium Truck | Puneeth | thor | 100 | 10000 | 17050 | COMPLETED |
| 2 | Kollam | Trivandrum | TBD | Large Truck | Suresh | Sura | 74 | 1000 | 11650 | COMPLETED |
| 3 | Chennai | Trivandrum | TBD | Large Truck | Biju | sura | 500 | 500 | 75300 | IN_PROGRESS |
| 6 | AA | BB | TBD | Small Truck | Nelson | Giri | 500 | 220 | 45160 | CANCELLED |
| 7 | LL | KK | TBD | Small Truck | JJ | girindra | 10 | 5 | 952.5 | COMPLETED |
| 8 | Chennai | Trivandrum | TBD | Large Truck | Sanat | Shyam | 500 | 300 | 75200 | COMPLETED |
| 9 | Kollam | Trivandrum | TBD | Small Truck | Snat | B | 73 | 100 | 6670 | IN_PROGRESS |
| 10 | Kollam | Trivandrum | TBD | Large Truck | sugu | NULL | 73 | 500 | 11250 | PENDING |
| 11 | Trivandrum | Kollam | TBD | Small Truck | Sugunan | AAC | 65 | 1000 | 6400 | COMPLETED |
| 13 | kollam | palakkad | TBD | Large Truck | NN | AA1 | 700 | 400000 | 305050 | IN_PROGRESS |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |